

Microbiome Basics – For AI and Non-Bio Experts

1. Microbiome

The **total collection of microorganisms** (bacteria, archaea, fungi, viruses) and their **genetic material** in a specific environment, like the human gut, skin, or lungs.

- In humans, the microbiome plays essential roles in:
 - digestion and metabolism
 - immune system regulation
 - protection against pathogens
 - possibly influencing mood, cognition, and disease risk

2. Microbiota vs. Microbiome

- **Microbiota** = the **organisms themselves** (e.g., all bacteria in your gut)
- **Microbiome** = **microbiota + their genes + their environment + their functions**

3. Host–Microbiome Interaction

The dynamic, reciprocal relationship between a human (or other host organism) and its microbial community is critical to health and disease.

4. Dysbiosis

An **imbalance** in the microbiome, often marked by loss of beneficial microbes, expansion of harmful microbes, or reduced diversity, linked to many diseases (e.g., IBD, obesity, cancer).

5. Commensal vs. Pathogen

- **Commensal microbes** coexist without harming the host (often beneficial).
- **Pathogens** are disease-causing organisms.
 - Some microbes can shift roles depending on the environment (opportunistic).

6. Human Microbiome Project (HMP)

A large NIH initiative that characterized the baseline microbiome of healthy individuals, establishing reference datasets still used today.

7. DNA RNA protein - central dogma (which is wrong)

16S rRNA Sequencing vs. Shotgun Metagenomics

- **16S rRNA Sequencing:**
 - Targets one gene (16S rRNA) to identify **bacteria and archaea**.
 - Cheaper, more standardized, lower resolution.
- **Shotgun Metagenomics:**
 - Sequences **all DNA** in a sample.
 - Captures broader microbial diversity (bacteria, fungi, viruses) and **functional genes**.
 - More data, more cost, more complex.

8. Microbial Community Profiling

Describing the composition (what taxa are there) and relative abundance of microbes in a sample.

9. Prebiotics, Probiotics, Postbiotics

- **Prebiotics:** nutrients that feed beneficial microbes
- **Probiotics:** live beneficial microbes
- **Postbiotics:** beneficial compounds produced by microbes (e.g., SCFAs like butyrate)

10. Multi-Omics Integration

In microbiome research, combining:

- **metagenomics** (what organisms are there),
- **metatranscriptomics** (what they're expressing),
- **metabolomics** (what they're producing),
- **host data** (phenotype, genetics, immune markers)

This integrative approach often leverages AI/ML to find complex biological patterns.

What is microbiome taxa?

In microbiome analysis, "taxa" refers to the different groups of microorganisms, such as bacteria, fungi, or viruses, that are present in a sample. Microbiome studies involve identifying and quantifying these taxa to understand the composition and function of the microbial community. These taxa are organized into a taxonomic hierarchy, with levels like domain, kingdom, phylum, class, order, family, genus, and species.

Here's a more detailed explanation:

1. What are Taxa?

- Taxa are defined groups of organisms, essentially the classification system for living things.
- In microbiome studies, taxa refer to the various types of microorganisms present in a sample, such as different species of bacteria.

2. Why are Taxa Important in Microbiome Studies?

Identifying the Community:

Taxa identification helps determine which microorganisms are present in a specific habitat or host, providing insights into the composition of the microbiome.

Understanding Function:

By identifying the different taxa, researchers can infer the potential functions of the microbial community, as certain taxa are known to perform specific metabolic or ecological roles.

Detecting Changes:

Changes in the relative abundance of different taxa can indicate shifts in the microbial community, potentially related to health conditions or environmental changes.

Targeting Intervention:

Understanding the composition of the microbiome can help researchers identify potential targets for interventions, such as therapeutic strategies or biopesticides.

3. Examples of Taxa in Microbiome Studies

- **Bacterial Taxa:** Commonly studied in the gut, skin, and other environments.
- **Fungal Taxa:** Can be found in various environments, including the gut and soil.
- **Viral Taxa:** While less frequently studied, viruses also play a role in the microbiome.

4. How are Taxa Analyzed in Microbiome Studies?

- **16S rRNA Gene Sequencing:** A common method for identifying bacterial taxa.

- **Metagenomics:** Sequencing of all DNA from a sample to identify all taxa, not just bacteria.
- **Statistical Analysis:** Data from sequencing is analyzed to determine relative abundance, differential abundance, and other metrics.

5. Taxa and the Core Microbiome

- The core microbiome refers to the set of taxa that are consistently present and abundant in a particular environment or host.
- Rare taxa, which are present at lower abundances, can also play important roles in maintaining the stability and diversity of the microbiome.

In summary, taxa are fundamental to understanding the composition and function of the microbiome, and analyzing their relative abundances and distributions provides valuable insights into microbial communities and their interactions with their environment.

OTU (Operational Taxonomic Unit):

An OTU is a cluster of DNA sequences that are grouped together based on similarity, typically at 97% identity. It serves as a proxy for a microbial "species" in early microbiome studies. OTUs help simplify analysis by grouping closely related sequences into a single unit, but they can lump together biologically distinct organisms.

👉 Think of OTUs as "bins" of similar microbes, defined by arbitrary similarity thresholds.

♦ ASV (Amplicon Sequence Variant):

An ASV is a **precisely resolved DNA sequence** variant inferred from amplicon data (like 16S rRNA sequencing). Tools like **DADA2** or **Deblur** denoise raw reads to distinguish even single-nucleotide differences, providing higher resolution than OTUs.

👉 ASVs are exact sequence variants, reproducible across datasets, and more biologically meaningful for tracking fine-scale microbial changes.

Microbial taxonomy is the science of classifying, identifying, and naming microorganisms, including bacteria, archaea, fungi, and viruses. It involves grouping microbes based on shared characteristics, both phenotypic (observable traits) and genotypic (genetic makeup). This helps us understand microbial diversity, ecology, and develop strategies for combating infections.

Key aspects of microbial taxonomy include:

Classification:

Grouping microbes into hierarchical ranks (e.g., domain, kingdom, phylum, class, order, family, genus, species) based on evolutionary relationships and/or shared characteristics.

Identification:

Determining the specific species or group of microorganisms from a sample.

Nomenclature:

Assigning standardized names to microorganisms, following established rules.

Characterization:

Studying the traits of microorganisms to understand their biology and function.

The Importance of Microbial Taxonomy:

Understanding Microbial Diversity:

Taxonomy helps us understand the vast array of microorganisms and their roles in various ecosystems.

Disease Diagnosis and Treatment:

Accurate identification of pathogens is crucial for diagnosing and treating infectious diseases.

Biotechnology and Research:

Microbial taxonomy provides a framework for studying and utilizing microorganisms in various applications, such as biotechnology and environmental science.

Evolution of Microbial Taxonomy:

- Early systems focused on observable characteristics (morphology, physiology).
- Molecular biology advancements led to a shift towards genomic-based taxonomy, using DNA sequences to classify microbes.
- The three-domain system (Bacteria, Archaea, Eukarya) is now widely used.